

Calculus II

Integrals of the form $\int \frac{Ax + B}{ax^2 + bx + c} dx$,
denominator has no real roots

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2019

Linear substitutions leading to blocks IIa and IIIa

Building block IIa: $\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2) + C.$

Building block IIIa: $\int \frac{1}{1+x^2} dx = \arctan x + C.$

- Let $ax^2 + bx + c$ have no real roots.
- We can find p, q so that the linear substitution $u = px + q$ transforms the quadratic to:

$$ax^2 + bx + c = r(u^2 + 1)$$

(where r is some number to be determined).

- To find p, q , we complete the square.
- In this way, integrals of the form $\int \frac{Ax + B}{ax^2 + bx + c} dx$ are transformed to combinations of building blocks IIa and IIIa.
- We show examples; the general case is analogous and we leave it to the student.

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Example

No real roots $\Rightarrow$ complete the square. Let $u = x + \frac{1}{2}$, let $z = \frac{2u}{\sqrt{3}}.$

$$\begin{aligned}
 \int \frac{x}{x^2 + x + 1} dx &= \int \frac{x}{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} dx \\
 &= \int \frac{x + \frac{1}{2} - \frac{1}{2}}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} d\left(x + \frac{1}{2}\right) \\
 &= \int \frac{u - \frac{1}{2}}{u^2 + \frac{3}{4}} du \\
 &= \int \frac{u}{u^2 + \frac{3}{4}} du - \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} du
 \end{aligned}$$

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$$\begin{aligned} \int \frac{x}{x^2 + x + 1} dx &= \int \frac{u}{u^2 + \frac{3}{4}} du - \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} du \\ \int \frac{1}{u^2 + \frac{3}{4}} du &= \int \frac{1}{\frac{3}{4} \left(\frac{4}{3} u^2 + 1 \right)} du \\ &= \int \frac{1}{\frac{3}{4} \left(\left(\frac{2u}{\sqrt{3}} \right)^2 + 1 \right)} \frac{\sqrt{3}}{2} d\left(\frac{2u}{\sqrt{3}} \right) \\ &= \frac{2\sqrt{3}}{3} \int \frac{1}{z^2 + 1} dz = \frac{2\sqrt{3}}{3} \arctan z + C \end{aligned}$$

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$$\begin{aligned} \int \frac{x}{x^2 + x + 1} dx &= \int \frac{u}{u^2 + \frac{3}{4}} du - \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} du \\ &= \int \frac{u}{u^2 + \frac{3}{4}} du - \frac{1}{2} \frac{2\sqrt{3}}{3} \arctan z + C \end{aligned}$$

$$\begin{aligned} \int \frac{u}{u^2 + \frac{3}{4}} du &= \int \frac{1}{u^2 + \frac{3}{4}} d\left(\frac{u^2}{2}\right) \\ &= \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} d\left(u^2 + \frac{3}{4}\right) = \frac{1}{2} \ln\left(u^2 + \frac{3}{4}\right) + C \end{aligned}$$

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$$\begin{aligned}
 \int \frac{x}{x^2 + x + 1} dx &= \int \frac{u}{u^2 + \frac{3}{4}} du - \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} du \\
 &= \frac{1}{2} \ln \left(u^2 + \frac{3}{4} \right) - \frac{1}{2} \frac{2\sqrt{3}}{3} \arctan z + C \\
 &= \frac{1}{2} \ln \left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right) - \frac{\sqrt{3}}{3} \arctan \left(\frac{2u}{\sqrt{3}} \right) + C \\
 &= \frac{1}{2} \ln \left(x^2 + x + 1 \right) - \frac{\sqrt{3}}{3} \arctan \left(\frac{2x+1}{\sqrt{3}} \right) + C
 \end{aligned}$$